

Study Report: Hands-On Machine Learning - ML / AI Engineer

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Family: ML / AI Engineer
Level: Intermediate
Chapters: 7
Source: Reference: Hands-On Machine Learning - Aurélien Géron

Official sources and free libraries

- <https://www.oreilly.com/library/view/hands-on-machine-learning/9781098125974/>

Summary

Study report for **Hands-On Machine Learning** by Aurélien Géron (Intermediate level) mapped to the ML / AI Engineer role. End-to-end ML with scikit-learn, TensorFlow, and deployment.

Chapters

Track: Book-Intro

Chapter 1: About Hands-On Machine Learning [Intermediate]

Chapter focus: About Hands-On Machine Learning** *(Level: Intermediate)

This study report summarizes **Hands-On Machine Learning** by Aurélien Géron for the ML / AI Engineer role. The resource is rated Intermediate level. End-to-end ML with scikit-learn, TensorFlow, and deployment. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Hands-On Machine Learning
# Author: Aurélien Géron
# Role: ML / AI Engineer
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Hands-On Machine Learning.

When to use it

When learning ML / AI Engineer skills at Intermediate level.

Where to use it

ML / AI Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role** *(Level: Intermediate)

ML / AI Engineer professionals use ideas from Hands-On Machine Learning to solve real workplace problems. End-to-end ML with scikit-learn, TensorFlow, and deployment. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: ML / AI Engineer

Book focus: End-to-end ML with scikit-learn, TensorFlow, and deployment.

Recommended level: Intermediate

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

ML / AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered** *(Level: Intermediate)

The main topics in Hands-On Machine Learning include practical concepts described as: End-to-end ML with scikit-learn, TensorFlow, and deployment. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in ML / AI Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Hyperparameter Tuning with Optuna [Intermediate]

Chapter focus: Hyperparameter Tuning with Optuna** *(Level: Intermediate)

Grid search explodes combinatorially; Optuna uses TPE sampler for efficient search. Log trials to MLflow; prune unpromising runs early.

Code Reference:

```
import optuna

def objective(trial):
    c = trial.suggest_float('C', 1e-3, 10, log=True)
    clf = LogisticRegression(C=c)
    return cross_val_score(clf, X, y, cv=3, scoring='roc_auc').mean()

study = optuna.create_study(direction='maximize')
```

```
study.optimize(objective, n_trials=50)
```

What it shows: suggest_float with log scale explores regularization efficiently.

Topic guide

What it actually is

Automated tuning finds better models with less manual guessing.

When to use it

After baseline model plateaus.

Where to use it

Competitions, production retrains, and research spikes.

Real use example

Optuna finds C=0.3 beats default - AUC +0.02 in 50 trials.

Chapter 5: Applied: MLflow Tracking and Model Registry [Intermediate]

Chapter focus: MLflow Tracking and Model Registry** *(Level: Intermediate)

mlflow.start_run logs parameters, metrics, artifacts. Registry versions models with Staging/Production stages and approval gates.

Code Reference:

```
import mlflow
with mlflow.start_run():
    mlflow.log_param('n_estimators', 200)
    mlflow.log_metric('auc', 0.91)
    mlflow.sklearn.log_model(pipe, 'model')
```

What it shows: log_model packages conda env and sklearn flavor for reproducible deploy.

Topic guide

What it actually is

MLflow is the experiment ledger and model lifecycle hub.

When to use it

Any team running more than ad-hoc notebook training.

Where to use it

MLOps platforms on AWS SageMaker, Databricks, or self-hosted.

Real use example

Champion model promoted to Production after CI integration tests pass.

Chapter 6: Applied: Imbalanced Classes and Threshold Tuning [Intermediate]

Chapter focus: Imbalanced Classes and Threshold Tuning** *(Level: Intermediate)

Use `class_weight`, SMOTE cautiously, or adjust decision threshold. Optimize F-beta or cost-sensitive metric - not default 0.5 threshold.

Code Reference:

```
from sklearn.metrics import precision_recall_curve
prec, rec, thresholds = precision_recall_curve(y_test, y_scores)
# pick threshold where recall >= 0.85
```

What it shows: PR curve reveals threshold tradeoffs ROC hides on imbalance.

Topic guide

What it actually is

Imbalance techniques align models with rare-event business goals.

When to use it

When positive class is rare and default 0.5 threshold misses business targets.

Where to use it

Fraud, disease screening, and critical churn saves.

Real use example

Payment fraud team sets threshold for 90% recall accepting 40% precision.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice *(Level: Intermediate)**

Finish Hands-On Machine Learning with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to ML / AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.